Das p-Bit als geometrischer Kipppunkt

Von der ξ-Längenhierarchie zur Torsionsbarriere

und die Auflösung der stochastischen Interpretation

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https://github.com/jpascher/T0-Time-Mass-Duality

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.1 Introduction: The Operating Point Between Bit and Qubit

Doc. 178 established the stability of spin as macroscopic evidence for discrete torsion configurations. The permanent magnet shows: spin-up and spin-down are geometrically distinguished states, robust over years without cooling. The classical bit uses precisely this regime — a deep potential well $U \gg k_B T$.

The qubit operates at the opposite extreme: metastable torsion, coherence in the millisecond range (Doc. 174), enforced by extreme cooling to mK temperatures.

The **p-bit** (probabilistic bit) occupies the conceptually hitherto unexplored intermediate range: a magnetic nanostructure on the geometric scale $L_8 \approx 22$ nm of the ξ-hierarchy, at which the torsion barrier becomes thermally surmountable.

The central thesis of this document

The thermal noise of the p-bit is not fundamental randomness. It is the macroscopically unresolved sampling of deterministic torsion configurations. The apparent stochasticity is epistemic — the transitions follow the sub-Planck-scale geometry operating on the timescale $t_0 = t_P/7500$.
The physical barrier is given materially by the Stoner-Wohlfarth energy $U = K_u \cdot V$. The FFGFT identifies: (a) **where** natural p-bit candidates are geometrically to be found (scale L_N), and (b) **how** the apparent randomness is to be understood physically.

.2 The ξ-Length Hierarchy (Review of Doc. 173)

Doc. 173 defines the geometric scaling ladder of the FFGFT via the parameter $\xi_{\text{par}} = 4/30000 \approx 1{,}333 \times 10^{-4}$:

$$L_N = L_0 \cdot \left(\frac{1}{\xi_{\text{par}}}\right)^N, \qquad L_0 = \xi_{\text{par}} \cdot \ell_P, \tag{1}$$

where $\ell_P = 1{,}616 \times 10^{-35}$ m is the Planck length. Level $N = 8$ gives:

$$L_8 = \ell_P \cdot \xi_{\text{par}}^{-7} \approx 21{,}6\,\text{nm}. \tag{2}$$

This scale is identified in Doc. 173 as the universal geometric anchor: excitonic Bohr radius, fine structure constant, transistor threshold size — all at $N \approx 8{,}00$.

.3 From Length Scale to Characteristic Energy

Torsion field characteristic at scale L_N

Each geometric scale L_N is assigned a characteristic torsion energy in the FFGFT framework. The torsion field is a relativistic field object; the natural energy scale of a torsion mode at wavelength L_N is:

$$E_N \equiv \frac{\hbar c}{L_N} = \frac{\hbar c}{L_0}\xi_{\text{par}}^N = \frac{E_P}{\xi_{\text{par}}}\xi_{\text{par}}^N = E_P \cdot \xi_{\text{par}}^{N-1}, \tag{3}$$

where $E_P = \hbar c/\ell_P = \hbar/t_P \approx 1{,}956 \times 10^9$ J is the Planck energy. Since $\xi_{\text{par}} < 1$, E_N decreases strictly monotonically with increasing N; each level is a factor of $1/\xi_{\text{par}} = 7500$ smaller than the previous one.

Verification: $N = 1$ yields the Planck energy

For $N = 1$:

$$E_1 = E_P \cdot \xi_{\text{par}}^0 = E_P \approx 1{,}956 \times 10^9 \text{ J}. \tag{4}$$

The Planck energy is the torsion energy of the Planck level — not a free parameter, but a geometric consequence of the hierarchy.

Complete energy hierarchy and the p-bit regime

N	L_N	$E_N = \hbar c/L_N$	$E_N/k_B T_{300}$	Physical domain
0	$2{,}16 \times 10^{-39}$ m	$1{,}47 \times 10^{13}$ J	$3{,}5 \times 10^{33}$	sub-Planck (L_0)
1	$1{,}62 \times 10^{-35}$ m	$1{,}96 \times 10^{9}$ J	$4{,}7 \times 10^{29}$	Planck scale; $E_1 = E_P$
2	$1{,}21 \times 10^{-31}$ m	$2{,}61 \times 10^{5}$ J	$6{,}3 \times 10^{25}$	—
3	$9{,}09 \times 10^{-28}$ m	$3{,}48 \times 10^{1}$ J	$8{,}4 \times 10^{21}$	—
4–7	...	...	10^{18}–10^{6}	—
8	**$2{,}16 \times 10^{-8}$ m**	**$1{,}47 \times 10^{-18}$ J**	**354**	**Exciton, Bohr radius, transistor threshold**
9	$1{,}62 \times 10^{-4}$ m	$1{,}95 \times 10^{-22}$ J	**0,047**	$0{,}16$ mm
10	$1{,}21 \times 10^{0}$ m	$2{,}61 \times 10^{-26}$ J	6×10^{-6}	~ 1 m

The gap across the p-bit regime

The p-bit operating regime $E_N \approx k_B T$ at $T = 300$ K lies at $N_{\text{opt}} \approx 8{,}66$ — *between* the integer levels $N = 8$ and $N = 9$. The step factor is $1/\xi_{\text{par}} = 7500$: between E_8 ($354\,k_B T$) and E_9 ($0{,}047\,k_B T$) there is no integer level value that hits $k_B T(300\,\text{K})$.

Consequence: The ξ-energy hierarchy E_N does *not* directly fix the barrier height of the p-bit. It determines the **geometric scale** of the nanostructure at which p-bit-capable resonators naturally occur.

For cryogenic p-bits, however: $E_9 \approx k_B T$ at $T^* \approx 14$ K. At this temperature a macroscopic resonator of dimension $L_9 \approx 0{,}16$ mm lies exactly on a hierarchy level.

.4 The Physical Barrier: Stoner-Wohlfarth Energy

Derivation

A magnetic nanoresonator with uniaxial anisotropy possesses an energy barrier between spin-up and spin-down, given in the Stoner-Wohlfarth model [4] by:

$$U_{\text{sw}} = K_u \cdot V \tag{5}$$

Here K_u is the uniaxial anisotropy constant (in J/m^3, material-specific) and V is the volume of the magnetic free layer.

In the FFGFT framework, U_{sw} is the macroscopic manifestation of the torsion barrier between the two stable configurations that Doc. 178 identifies as geometrically distinguished states.

For a spherical nanoresonator with diameter D:

$$U_{\text{sw}}(D) = K_u \cdot \frac{\pi}{6} D^3. \tag{6}$$

The condition for p-bit operation $U_{\text{sw}} \approx k_B T$ fixes a specific diameter D^* for a given K_u:

$$D^* = \left(\frac{6\,k_B T}{\pi K_u}\right)^{1/3}. \tag{7}$$

Numerical values and comparison with L_8

For superparamagnetic nanoparticles with $K_u \approx 10^4$ J/m^3:

D (nm)	U_{sw} (meV)	$U_{\mathrm{sw}}/k_B T_{300}$	Regime
5	4,1	0,16	
8	16,7	0,65	p-bit
10	32,7	1,26	p-bit
12	56,5	2,18	p-bit
15	110	4,3	p-bit
20	261	10,1	p-bit
25	511	19,8	

The p-bit operating range ($U/k_B T \approx 1$–10) lies at $D \approx 8$–20 nm — exactly in the range of the ξ-hierarchy level $N = 8$ with $L_8 \approx 22$ nm.

Geometric prediction: $N = 8$ as natural p-bit anchor

The FFGFT predicts: magnetic nanoresonators on the ξ-level $N \approx 8$ are natural p-bit candidates at room temperature. Not because $E_8 \approx k_B T$ (that is not the case: $E_8 = 354\, k_B T$), but because $L_8 \approx 22$ nm is the geometric scale at which the Stoner-Wohlfarth volume with typical material values falls into the p-bit regime.
This coincidence is not accidental in the FFGFT: $N = 8$ is the universal scale of magnetic single resonators (Doc. 173: minimal resonator, Doc. 178: permanent magnet as collective resonator at $N \approx 8$).

.5 The Kramers Switching Rate in the FFGFT Framework

Standard formula

The thermally activated switching rate between torsion configurations follows the Kramers formula [5, 6]:

$$\Gamma = \frac{\omega_0\, \omega_b}{2\pi\gamma} \exp\left(-\frac{U_{\mathrm{sw}}}{k_B T}\right), \tag{8}$$

where ω_0 is the angular frequency at the potential minimum, ω_b at the barrier peak and γ is the (Gilbert) damping rate.

FFGFT interpretation: determinism instead of randomness

In the standard interpretation, Γ is a stochastic rate: each transition is a random, uncorrelated event.

FFGFT reinterpretation of the Kramers rate

In the FFGFT, equation (8) is **formally identical** — but its physical meaning changes:
Γ is the **time-averaged frequency of deterministic torsion transitions**. Each individual transition follows a precise path in the sub-Planck-scale geometry of the torsion field, which is determined on the timescale $t_0 = t_P/7500$.
The macroscopic measurement — temporal resolution $\Delta t \gg t_0$ — averages over $\Delta t/t_0 \gg 1$ microstates and creates the appearance of randomness. The randomness is **epistemic**, not ontological.
This is the direct transfer of the FFGFT position established in Doc. 178 (epistemic ignorance rather than genuine superposition) to the thermally fluctuating p-bit system.

Why no modified prefactor

A numerical FFGFT correction to the Kramers rate would require the structural parameter $k_BTt_0/\hbar$:

$$\frac{k_BTt_0}{\hbar} = \frac{k_B \cdot 300\,\mathrm{K}}{\hbar/t_0} \approx 2{,}82 \times 10^{-34}. \tag{9}$$

This value is $\ll 1$ at all technically accessible temperatures. A correction to the Kramers prefactor by the FFGFT is not experimentally detectable at room temperature.

Where the FFGFT contribution really lies

1. **Geometric scale:** $D \sim L_8 \approx 22\,\mathrm{nm}$ as natural p-bit anchor — verifiable materials-science prediction.
2. **Interpretation:** The switching statistics are deterministically structured on the t_0 scale; the macroscopically observed stochasticity is a projection.

3. **No new terms:** The Kramers rate (8) remains unchanged. The FFGFT contribution is interpretive.

.6 The Three-Regime Picture: Bit, p-Bit, Qubit

Type	Barrier	Temp.	Timescale
Class. bit	$U \gg k_B T$	300 K	Years
p-bit	$U \approx k_B T$	300 K	ns–μs
Qubit	$U \gg k_B T$ (prep.)	mK	μs–ms

Type	ξ-level	FFGFT: random?
Class. bit	$N \approx 8$, deep	No; stable torsion well
p-bit	$D \sim L_8 \approx 22\,\text{nm}$	No; deterministic
Qubit	$N \approx 8$, metastable	No; epistemic (Doc. 178)

All three types reside on the same geometric scale $N \approx 8$. The difference is the operating point on the potential curve: deep in the minimum (bit), at the tipping point (p-bit), metastably prepared (qubit).

.7 Landauer's Principle

Landauer's principle [7] applies to the p-bit unchanged:

$$E_{\text{diss}}^{\text{min}} = k_B T \ln 2. \tag{10}$$

The p-bit switches between two torsion configurations; the state space is binary. The ξ-geometry determines the *dynamics* (scale, switching rate), not the *state space size*.

No modified Landauer bound in the FFGFT

A ξ-dependent Landauer bound would require the state space of the p-bit to have more than two elements, or that information erasure uses geometrically non-equivalent paths. Neither is the case in a standard p-bit. The standard bound $k_B T \ln 2$ applies.

.8 Experimental Realisations

- **Magnetic Tunnel Junctions (MTJ) [2]:** CoFeB free layers, $K_u \sim 10^4$–10^5 J/m^3, lateral dimension $\sim$ 20–100 nm — directly at L_8. Switching times in the nanosecond range.
- **Superparamagnetic nanoparticles:** Fe_3O_4, γ-Fe_2O_3 with $D \approx$ 10–20 nm show spontaneous thermal switching at 300 K in the L_8 regime.
- **CMOS-based p-bits [3]:** Ring oscillators near metastability. No direct torsion field connection — realise the logical operating point by other means.

Verifiable FFGFT prediction

The FFGFT predicts a **geometrically distinguished dimension** $D^* \approx L_8 \approx 22$ nm for optimal p-bit quality — material-independent, as long as K_u scales accordingly. A systematic measurement of switching rate, noise colour and reproducibility across different nanoparticle sizes should show an optimum at $D \approx L_8$.

.9 Classification within the FFGFT Document Series

Doc.	Topic	Relation to Doc. 184
173	ξ-hierarchy, quantum dot, minimal resonator	Foundation: L_N, E_N derivation, $N = 8$ anchor
174	Reading and writing spin	Techniques; decoherence times
178	Spin stability, permanent magnet	Epistemic randomness; torsion well vs. tipping point
184	**p-bit**	**Tipping point: same scale $N = 8$, different operating point**

.10 Summary

Conclusion: p-bit in the FFGFT framework

1. The ξ-energy hierarchy is cleanly derivable.
$E_N = \hbar c/L_N = E_P \cdot \xi_{\text{par}}^{N-1}$, with $E_1 = E_P$. The levels jump by a factor of 7500 — the p-bit regime at 300 K lies between $N = 8$ and $N = 9$.

2. The barrier is Stoner-Wohlfarth: $U = K_u \cdot V$.
Adjustable by materials engineering, not quantised by E_N. p-bit regime at $D \approx 8$–20 nm ($K_u \sim 10^4$ J/m^3).

3. The ξ-reference is the geometric scale $L_8 \approx 22$ nm.
This is the universal resonator scale $N = 8$ from Doc. 173. Natural p-bit candidates lie on this level.

4. The Kramers rate remains formally unchanged.
FFGFT contribution: determinism of transitions on the t_0 scale — the stochasticity is epistemic, not ontological.

5. The Landauer bound applies unchanged.
$E_{\text{diss}}^{\text{min}} = k_B T \ln 2$ — binary state space, no ξ modification.

The p-bit is the geometric tipping point between permanent magnet (Doc. 178) and qubit (Doc. 174) — on the same ξ-level $N = 8$.

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